$$W_2 = \frac{15}{16}E^2C \qquad (16)$$

$$\frac{l}{v_n} = n\frac{T}{2} \quad n = 1,2,\cdots \qquad (1)$$

$$v_n = \frac{2l}{nT} \qquad n = 1,2,\cdots \qquad (2)$$

$$\Delta t_{n1} = \frac{1}{2}nT \qquad (3)$$

$$\Delta t_{n2} = \frac{l}{2v_n} \qquad (4)$$

$$t_n = \Delta t_{n1} + \Delta t_{n2} = \frac{3}{4}nT \qquad (5)$$

$$t_n \leqq \frac{10}{4}T \qquad (6)$$

$$n_{\max} = 3 \qquad (7)$$

$$E_n = \frac{1}{2}mv_n^2 + E_0\sin\left[\frac{2\pi}{T}\left(\frac{3n}{4}T\right) + \frac{\pi}{4}\right]$$
$$n = 1,2,3 \qquad (8)$$